

#1

[A, E, 1], [A, H, 1], [E, C, 1], [H, D, 1], [C, F, 1], [D, E, 1]
[D, A, 1], [F, D, 1], [F, E, 1], [G, E, 1], [G, B, 1], [B, A, 1]

#2

A:0, B:1, C:2, D:3, E:4, F:5, G:6, H:7

A = E, H

Vertex **A** is visited

Current array: [0, -1, -1, -1, -1, -1, -1, -1]

Vertex **E** and **H** enqueued, Current queue: [A, E, H]

Vertex **A** is dequeued, Current queue: [E, H]

E = C

Vertex **E** visited

Current array: [0, 4, -1, -1, -1, -1, -1, -1]

Vertex **C** is enqueued Current queue: [E, H, C]

Vertex **E** is dequeued, Current queue: [H, C]

H = D

Vertex **H** is visited

Current array: [0, 4, 7, -1, -1, -1, -1, -1]

Vertex **D** is enqueued, Current queue: [H, C, D]

Vertex **H** is dequeued, Current queue: [C, D]

C = F, G

Vertex **C** is visited

Current array: [0, 4, 7, 2, -1, -1, -1, -1]

Vertex **F** and **G** are enqueued, Current queue: [C, D, F, G]

Vertex **C** is dequeued, Current queue: [D, F, G]

D = E, A

Vertex **D** is visited

Current array: [0, 4, 7, 2, 3, -1, -1, -1]

Vertex **D** is dequeued, Current queue: [F, G]

Vertex **E** and **A** already visited

F = D, E

Vertex **F** is visited

Current array: [0, 4, 7, 2, 3, 5, -1, -1]

Vertex **F** is dequeued, Current queue: [G]

Vertex **D** and **E** already visited

G = E, B

Vertex **G** is visited

Current array: [0, 4, 7, 2, 3, 5, 6, -1]

Vertex **B** is enqueued, Current queue: [G, B]

Vertex **G** is dequeued, Current queue: [B]

Vertex **E** already visited

B = A

Vertex **B** is visited

Current array: [0, 4, 7, 2, 3, 5, 6, 1]

Vertex **B** is dequeued, Current queue: []

Vertex **A** already visited

Final Array: [A:0, E:4, H:7, C:2, D:3, F:5, G:6, B:1]